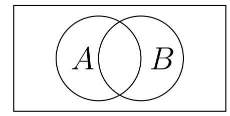
Homework 2

1. Given that $\{x\} \cup (3,7] = [3,7]$, find x.



2. If A and B are the sets shown in the picture, shade $(A \setminus B) \cup (B \setminus A)$.

3. Two sets A and B are said to be disjoint if they have no elements in common, i.e., $A \cap B = \varnothing$. Among the sets below, find all pairs of disjoint sets.

   $A = \{1,2\}$ $\quad B = \{2,4,6\}$ $\quad C = \{4,5,6,7\}$ $\quad D = \{3,5,8\}$ $\quad E = \{1,4\}$

4. Show that we can have $A \cap B = A \cap C$ without $B = C$.

5. If the experiment consists of tossing a coin and rolling a die, find the sample space.

6. Suppose 50 students are polled to see whether or not they have studied French (F) or German (G) yielding the following data: 25 studied French, 20 studied German, 5 studied both. Find the number of the students who studied: (a) only French, (b) French or German, (c) neither language.

7. Out of 10 student candidates, in how many ways can 3 students be selected to continue their studies abroad in Germany?

8. Three students are selected from 10 candidates and assigned to three universities in Germany—TUM, Heidelberg University, and RWTH Aachen University; in how many ways can this be done?

9. Find the number of ways 9 toys may be divided among four children if the youngest is to receive 3 toys and each of the others 2 toys.

10. A box contains 5 black balls and 7 white balls. Three balls are drawn at random. What is the probability that two of them are white and one is black?

11. A box contains 6 black balls, 7 white balls, and 8 blue balls. 5 balls are drawn at random. What is the probability that exactly two of them are white and exactly one is black?

12. Find the probability that a 5-card hand drawn from a deck of 52 playing cards contains: (a) exactly one ace, (b) exactly 3 aces, (c) exactly 2 kings, (d) exactly two kings and two queens, (e) exactly two cards of the same suit. (There are 4 suits: spades, hearts, diamonds, clubs)

13. In how many ways can 3 novels, 2 mathematics books, and 1 chemistry book be arranged on a bookshelf if

    (a) the books can be arranged in any order?
    (b) the mathematics books must be together and the novels must be together?
    (c) the novels must be together, but the other books can be arranged in any order?

14. Alan, Bob, and eight other students (10 students in total) are arranged in a row at random. Find the probability that Alan and Bob stand next to each other.